

Supporting information for: Estimating Atomic Contributions to Hydration and Binding Using Free Energy Perturbation

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Abstract

This supplementary gives the derivations for the AFEP formulae. And some additional details into the molecular dynamics simulations performed.

Appendix 1: Detailed Derivation to Reach Expanded Form

We start from the Zwanzig equation for the Helmholtz free energy difference ΔF_{AB} between two thermodynamic states A and B

$$\Delta F_{AB} = F_B - F_A = -\beta^{-1} \log \langle e^{-\beta(U_B - U_A)} \rangle_A, \quad (1)$$

The ensemble average in equation 1 can be written

$$\Delta F_{AB} = -\beta^{-1} \log \left[\frac{1}{Q_A} \int e^{-\beta(U_B(\vec{q}) - U_A(\vec{q}))} e^{-\beta U_A(\vec{q})} d\vec{q} \right] \quad (2)$$

where Q_A is the partition function of system A

$$Q_A = \int e^{-\beta U_A(\vec{q})} d\vec{q} \quad (3)$$

and $\vec{q}$ is a $3N$ dimensional vector, where there are N atoms in the system. We want to express this in terms of individual atoms, so we expand the potential energy in terms of these atoms

$$U_A(\vec{q}) = \sum_{k=1}^N u_{Ak}(q_k) \quad (4)$$

$$U_B(\vec{q}) = \sum_{k=1}^N u_{Bk}(q_k) \quad (5)$$

$$U_B(\vec{q}) - U_A(\vec{q}) = \sum_{k=1}^N \Delta u_k(q_k) \quad (6)$$

$$\Delta u_k = u_{Bk}(q_k) - u_{Ak}(q_k) \quad (7)$$

where the u_{Xk} is the potential energy function of atom k in system X . We still want to retain the reference to system A so we only expand the difference term

$$\Delta F_{AB} = -\beta^{-1} \log \left[\frac{1}{Q_A} \int e^{-\beta(\sum_{k=1}^N u_{Bk}(q_k) - u_{Ak}(q_k))} e^{-\beta U_A(\vec{q})} d\vec{q} \right] \quad (8)$$

Our approach is to expand the whole exponentiated sum in the equation

$$\Delta F_{AB} = -\beta^{-1} \log \left[\frac{1}{Q_A} \int e^{-\beta \sum_{k=1}^N \Delta u_k} e^{-\beta U_A(\vec{q})} d\vec{q} \right] \quad (9)$$

we can use the Maclaurin Series for $e^{-\alpha x}$ which gives

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = 1 - \beta \sum_{k=1}^N \Delta u_k + \frac{\beta^2}{2!} \left(\sum_{k=1}^N \Delta u_k \right)^2 - \frac{\beta^3}{3!} \left(\sum_{k=1}^N \Delta u_k \right)^3 + \dots \quad (10)$$

which when written in summation form is

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = \sum_{m=0}^{\infty} \frac{(-1)^m \beta^m}{m!} \left(\sum_{k=1}^N \Delta u_k \right)^m \quad (11)$$

The multinomial expansion identity is

$$(x_1 + x_2 + \dots + x_N)^m = \sum_{k_1+k_2+\dots+k_N=m} \frac{m!}{k_1!k_2!\dots k_N!} \prod_{t=1}^N x_t^{k_t} \quad (12)$$

where the sum is over all sets of indices $k_t \in [0, m]$ that sum to m . We can then write

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = \sum_{m=0}^{\infty} \frac{(-1)^m \beta^m}{m!} \sum_{k_1+k_2+\dots+k_N=m} \frac{m!}{k_1!k_2!\dots k_N!} \prod_{t=1}^N \Delta u_t^{k_t} \quad (13)$$

which can be reduced to

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = \sum_{m=0}^{\infty} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1!k_2!\dots k_N!} \prod_{t=1}^N \Delta u_t^{k_t} \right) \quad (14)$$

This is the point we implement the grouping scheme as mentioned in the main text, we introduce a weighting to each term such that terms with higher powers of the energy difference associated with atom a have a larger share in the term. It is clear that all of the summation terms in the above expansion are products of the Δu_t raised to different powers k_t with different coefficients. Say we are only interested in some atom with index a , then we can try and separate all the terms that contain atom a from the other terms and sum them together, we can call this sum $J(a)$. $J(a)$ will contain cross terms because the free energy is not directly separable in this manner. That is, terms which also contain the Δu_t

for atoms other than a . To deal with these cross terms we can partition (or weight) each term according to atoms present in them.

Here are two examples of the weighting scheme: A term that looks like $2\Delta u_1\Delta u_2$, is half ‘owned’ by atom 1 and half by atom 2, so the $J(1)$ and $J(2)$ terms will each contain a contribution of $\Delta u_1\Delta u_2$. A term that looks like $24\Delta u_1^2\Delta u_2\Delta u_3$, is $2/4$ ‘owned’ by atom 1, because of the power 2, and $1/4$ each by atoms 2 and 3, so the $J(1)$ term would have $12\Delta u_1^2\Delta u_2\Delta u_3$, and $J(2)$ and $J(3)$ each get $6\Delta u_1^2\Delta u_2\Delta u_3$. When written mathematically we prepare the exponential for the separation by removing the constant term of 1

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = \sum_{m=0}^{\infty} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1!k_2!\dots k_N!} \prod_{t=1}^N \Delta u_t^{k_t} \right) \quad (15)$$

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = 1 + \sum_{m=1}^{\infty} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1!k_2!\dots k_N!} \prod_{t=1}^N \Delta u_t^{k_t} \right) \quad (16)$$

this will become useful later for expanding the logarithm in the Zwanzig equation. Meanwhile we can enter the grouping rule above by introducing a factor of k_a/m and summing over all a giving

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = 1 + \sum_{a=1}^N \sum_{m=1}^{\infty} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1!k_2!\dots k_N!} \frac{k_a}{m} \prod_{t=1}^N \Delta u_t^{k_t} \right) \quad (17)$$

and then define the $J(a)$ function as

$$J(a) = \sum_{m=1}^{\infty} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1!k_2!\dots k_N!} \frac{k_a}{m} \prod_{t=1}^N \Delta u_t^{k_t} \right) \quad (18)$$

which gives

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = 1 + \sum_{k=1}^N J(k) \quad (19)$$

the reader is referred to Appendix 2 for the closed form of $J(a)$.

Appendix 2: Proof of the Closed Form for $J(a)$

We will prove that $J(a)$ as previously defined

$$J(a) = \sum_{m=1}^{\infty} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1! k_2! \dots k_N!} \frac{k_a}{m} \prod_{t=1}^N \Delta u_t^{k_t} \right) \quad (20)$$

can also be written as to

$$J(a) = \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} \left(e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right) \quad (21)$$

We can remove the m from inside the multinomial sum, and separate out the factorial of k_a , and the Δu_a term in the product to give

$$J(a) = \sum_{m=1}^{\infty} \frac{1}{m} \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1! k_2! \dots k_a! \dots k_N!} k_a \Delta u_a^{k_a} \prod_{t=1, t \neq a}^N \Delta u_t^{k_t} \right) \quad (22)$$

now we use the principle of distinguished element on k_a , if $k_a = 0$ the whole term is zero and otherwise we know k_a is in the range $1..m$, due to the conditions on the sum. If we take each case that k_a is in the range 1 to m and fully separate all terms out the inner part of the above sum becomes

$$\begin{aligned} & \left(\sum_{k_1+k_2+\dots+k_N=m} \frac{(-1)^m \beta^m}{k_1! k_2! \dots k_a! \dots k_N!} k_a \Delta u_a^{k_a} \prod_{t=1}^N \Delta u_t^{k_t} \right) = \\ & \sum_{k_a=1}^m \frac{k_a \Delta u_a^{k_a}}{k_a!} \left(\sum_{k_1+k_2+\dots+k_{a-1}+k_{a+1}+\dots+k_N=m-k_a} \frac{(-1)^m \beta^m}{k_1! k_2! \dots k_{a-1}! k_{a+1}! \dots k_N!} \prod_{t=1, t \neq a}^N \Delta u_t^{k_t} \right) \quad (23) \end{aligned}$$

we can divide the top and bottom by $(m - k_a)!$ and pull out the other m dependant terms

$$= (-1)^m \beta^m \sum_{k_a=1}^m \frac{1}{(m - k_a)!} \frac{k_a \Delta u_a^{k_a}}{k_a!} \left(\sum_{k_1+k_2+\dots+k_{a-1}+k_{a+1}+\dots+k_N=m-k_a} \frac{(m - k_a)!}{k_1! k_2! \dots k_{a-1}! k_{a+1}! \dots k_N!} \prod_{t=1, t \neq a}^N \Delta u_t^{k_t} \right) \quad (24)$$

we can simply relabel the index $k_a = s$

$$= (-1)^m \beta^m \sum_{s=1}^m \frac{1}{(m-s)!} \frac{s \Delta u_a^s}{s!} \left(\sum_{k_1+k_2+\dots+k_{a-1}+k_{a+1}+\dots+k_N=m-s} \frac{(m-s)!}{k_1!k_2!\dots k_{a-1}!k_{a+1}!\dots k_N!} \prod_{t=1, t \neq a}^N \Delta u_t^{k_t} \right) \quad (25)$$

we then compare and substitute this to the bracket expanded with the multinomial theorem

$$\left(\sum_{k=1, k \neq a}^N \Delta u_k \right)^{m-s} = \sum_{m-s=\sum_{i=1, i \neq a}^N k_i} \frac{(m-s)!}{\prod_{i=1, i \neq a}^N k_i!} \prod_{t=1, t \neq a}^N x_t^{k_t} \quad (26)$$

so we can write a new form for $J(a)$ as

$$J(a) = \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m}{m} \sum_{s=1}^m \frac{1}{(m-s)!} \frac{s \Delta u_a^s}{s!} \left(\sum_{k=1, k \neq a}^N \Delta u_k \right)^{m-s} \quad (27)$$

for convenience label the sum of the energy difference terms X

$$X = \sum_{k=1, k \neq a}^N \Delta u_k \quad (28)$$

then

$$J(a) = \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m}{m} \sum_{s=1}^m \frac{1}{(m-s)!} \frac{s \Delta u_a^s}{s!} X^{m-s} \quad (29)$$

also, as $s > 0$, we can take out a factor of Δu_a

$$J(a) = \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m \Delta u_a}{m} \sum_{s=1}^m \frac{X^{m-s}}{(m-s)!} \frac{\Delta u_a^{s-1}}{(s-1)!} \quad (30)$$

We have that

$$\sum_{s=1}^m \frac{X^{m-s}}{(m-s)!} \frac{\Delta u_a^{s-1}}{(s-1)!} = \frac{(X + \Delta u_a)^{m-1}}{(m-1)!} \quad (31)$$

which can be proved by a relabelling of indices from the binomial expansion

$$(a + b)^n = \sum_{k=0}^n \frac{n!}{k!(n-k)!} a^{n-k} b^k \quad (32)$$

under the mappings

$$a \rightarrow X, \quad b \rightarrow \Delta u_a, \quad n \rightarrow m-1, \quad k \rightarrow s-1 \quad (33)$$

using this identity gives

$$J(a) = \Delta u_a \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m}{m} \frac{(X + \Delta u_a)^{m-1}}{(m-1)!} \quad (34)$$

$$J(a) = \Delta u_a \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m}{m!} (X + \Delta u_a)^{m-1} \quad (35)$$

$$J(a) = \Delta u_a \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m}{m!} \frac{(X + \Delta u_a)^m}{(X + \Delta u_a)} \quad (36)$$

$$J(a) = \frac{\Delta u_a}{(X + \Delta u_a)} \sum_{m=1}^{\infty} \frac{(-1)^m \beta^m}{m!} (X + \Delta u_a)^m \quad (37)$$

$$J(a) = \frac{\Delta u_a}{(X + \Delta u_a)} (e^{-\beta(X + \Delta u_a)} - 1) \quad (38)$$

Finally we have that

$$X + \Delta u_a = \sum_{k=1}^N \Delta u_k \quad (39)$$

$$J(a) = \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} (e^{-\beta \sum_{k=1}^N \Delta u_k} - 1) \quad (40)$$

which was the desired result.

Appendix 3: Expansion of the Logarithm

From the Zwanzig equation, we have

$$\Delta F_{AB} = -\beta^{-1} \log \left[\frac{1}{Q_A} \int e^{-\beta \sum_{k=1}^N \Delta u_k} e^{-\beta U_A(\vec{q})} d\vec{q} \right] \quad (41)$$

and from the definition of $J(a)$

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = 1 + \sum_{k=1}^N J(k) \quad (42)$$

We can use this in the free energy equation

$$-\beta \Delta F_{AB} = \log \left[\frac{1}{Q_A} \int \left(1 + \sum_{k=1}^N J(k) \right) e^{-\beta U_A(\vec{q})} d\vec{q} \right] \quad (43)$$

which due to linearity is

$$-\beta \Delta F_{AB} = \log \left[1 + \sum_{k=1}^N \frac{1}{Q_A} \int J(k) e^{-\beta U_A(\vec{q})} d\vec{q} \right] \quad (44)$$

which is then the sum of the ensemble averages of the terms by definition

$$-\beta \Delta F_{AB} = \log \left[1 + \sum_{k=1}^N \langle J(k) \rangle \right] \quad (45)$$

There is not much that can be done with the logarithm of a sum, but as the 1 was conveniently left there from previous steps we can use another Taylor expansion on the logarithm this time

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (46)$$

$$-\beta \Delta F_{AB} = \sum_{k=1}^N \langle J(k) \rangle - \frac{1}{2} \left(\sum_{k=1}^N \langle J(k) \rangle \right)^2 + \frac{1}{3} \left(\sum_{k=1}^N \langle J(k) \rangle \right)^3 - \dots \quad (47)$$

or written in summation notation

$$-\beta\Delta F_{AB} = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \left(\sum_{k=1}^N \langle J(k) \rangle \right)^m \quad (48)$$

we can reuse the multinomial expansion rules for the powers of the sums of $J(a)$ as used in Appendix 1, and get

$$\Delta F_{AB}(a) = -\beta^{-1} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \sum_{k_1+k_2+\dots+k_N=m} \frac{m!}{k_1!k_2!\dots k_N!} \prod_{t=1}^N \langle J(t) \rangle^{k_t} \quad (49)$$

Similarly to the methods used in Appendix 1 we can define contributions

$$\Delta F_{AB} = \sum_{a=1}^N \Delta \mathcal{F}_{AB}(a) \quad (50)$$

where using the regrouping trick by including a factor of k_a/m gives

$$\Delta \mathcal{F}_{AB}(a) = -\beta^{-1} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \sum_{k_1+k_2+\dots+k_N=m} \frac{m!}{k_1!k_2!\dots k_N!} \frac{k_a}{m} \prod_{t=1}^N \langle J(t) \rangle^{k_t} \quad (51)$$

at this stage we have an atom-wise description of ΔF_{AB} , we just need to find a tractable closed form for the $\Delta \mathcal{F}_{AB}(a)$.

Appendix 4: Proof of Closed Form for $\Delta\mathcal{F}_{AB}(a)$

We will prove that

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \sum_{k_1+k_2+\dots+k_N=m} \frac{m!}{k_1!k_2!\dots k_N!} \frac{k_a}{m} \prod_{t=1}^N \langle J(t) \rangle^{k_t} \quad (52)$$

can also be written as

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \frac{\langle J(a) \rangle}{\sum_{k=1}^N \langle J(k) \rangle} \log \left(1 + \sum_{k=1}^N \langle J(k) \rangle \right) \quad (53)$$

once again as in Appendix 2, we will separate out all m dependant terms possible and all k_a terms, and divide top and bottom by $(m - k_a)!$, for compactness the condition in the multinomial sum can be rewritten in summation notation

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \frac{m!}{m} \sum_{k_a=1}^m \frac{k_a \langle J(a) \rangle^{k_a}}{k_a! (m - k_a)!} \sum_{m-k_a=\sum_{i=1, i \neq a}^N k_i} \frac{(m - k_a)!}{\prod_{i=1, i \neq a}^N k_i!} \prod_{t=1, t \neq a}^N \langle J(t) \rangle^{k_t} \quad (54)$$

as previously described in Appendix 2, labelling $k_a = s$ we may use the multinomial expansion identity

$$\left(\sum_{k=1, k \neq a}^N \langle J(k) \rangle \right)^{m-s} = \sum_{m-s=\sum_{i=1, i \neq a}^N k_i} \frac{(m-s)!}{\prod_{i=1, i \neq a}^N k_i!} \prod_{t=1, t \neq a}^N \langle J(t) \rangle^{k_t} \quad (55)$$

to give

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \frac{m!}{m} \sum_{s=1}^m \frac{\langle J(a) \rangle^s}{(s-1)! (m-s)!} \left(\sum_{k=1, k \neq a}^N \langle J(k) \rangle \right)^{m-s} \quad (56)$$

for convenience label the sum of terms as a separate variable

$$Y = \sum_{k=1, k \neq a}^N \langle J(k) \rangle \quad (57)$$

which gives

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \frac{m!}{m} \sum_{s=1}^m \frac{\langle J(a) \rangle^s Y^{m-s}}{(s-1)!(m-s)!} \quad (58)$$

as the summation index $s > 0$ we can remove a factor of $\langle J(a) \rangle$,

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \langle J(a) \rangle \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \frac{m!}{m} \sum_{s=1}^m \frac{\langle J(a) \rangle^{s-1} Y^{m-s}}{(s-1)!(m-s)!} \quad (59)$$

and then use the identity

$$\sum_{s=1}^m \frac{Y^{m-s}}{(m-s)!} \frac{\langle J(a) \rangle^{s-1}}{(s-1)!} = \frac{(Y + \langle J(a) \rangle)^{m-1}}{(m-1)!} \quad (60)$$

which again can be obtained from the binomial expansion

$$(a+b)^n = \sum_{k=0}^n \frac{n!}{k!(n-k)!} a^{n-k} b^k \quad (61)$$

under the mappings

$$a \rightarrow Y, \quad b \rightarrow \langle J(a) \rangle, \quad n \rightarrow m-1, \quad k \rightarrow s-1 \quad (62)$$

this gives us

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \langle J(a) \rangle \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \frac{m!}{m} \frac{(Y + \langle J(a) \rangle)^{m-1}}{(m-1)!} \quad (63)$$

$$\Delta\mathcal{F}_{AB}(a) = -\beta^{-1} \langle J(a) \rangle \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \frac{m!}{m!} \frac{(Y + \langle J(a) \rangle)^m}{(Y + \langle J(a) \rangle)} \quad (64)$$

$$\Delta\mathcal{F}_{AB}(a) = -\frac{\beta^{-1} \langle J(a) \rangle}{(Y + \langle J(a) \rangle)} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} (Y + \langle J(a) \rangle)^m \quad (65)$$

$$\Delta\mathcal{F}_{AB}(a) = -\frac{\beta^{-1} \langle J(a) \rangle}{(Y + \langle J(a) \rangle)} \log(1 + Y + \langle J(a) \rangle) \quad (66)$$

$$(67)$$

then we have that

$$Y + \langle J(a) \rangle = \sum_{k=1}^N \langle J(k) \rangle \quad (68)$$

giving

$$\Delta \mathcal{F}_{AB}(a) = -\beta^{-1} \frac{\langle J(a) \rangle}{\sum_{k=1}^N \langle J(k) \rangle} \log \left(1 + \sum_{k=1}^N \langle J(k) \rangle \right) \quad (69)$$

which was to be proved.

Appendix 5: Finalising the form of $\Delta \mathcal{F}_{AB}(a)$

Combining

$$\Delta \mathcal{F}_{AB}(a) = -\beta^{-1} \frac{\langle J(a) \rangle}{\sum_{k=1}^N \langle J(k) \rangle} \log \left(1 + \sum_{k=1}^N \langle J(k) \rangle \right) \quad (70)$$

and the closed form of $J(a)$

$$J(a) = \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} (e^{-\beta \sum_{k=1}^N \Delta u_k} - 1) \quad (71)$$

as well as the definition

$$e^{-\beta \sum_{k=1}^N \Delta u_k} = 1 + \sum_{k=1}^N J(k) \quad (72)$$

gives the final form

$$\Delta \mathcal{F}_{AB}(a) = -\beta^{-1} \frac{\left\langle \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} \left(e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right) \right\rangle}{\left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right\rangle} \log \left(1 + \left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right\rangle \right), \quad (73)$$

which is equal to

$$\Delta \mathcal{F}_{AB}(a) = -\beta^{-1} \frac{\left\langle \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} \left(e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right) \right\rangle}{\left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right\rangle} \log \left(\left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} \right\rangle \right), \quad (74)$$

which when considering the Zwanzig equation is

$$\Delta\mathcal{F}_{AB}(a) = \frac{\left\langle \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} \left(e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right) \right\rangle}{\left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right\rangle} \Delta F_{AB}. \quad (75)$$

This is the full atom-wise FEP expression. Summing over all a then gives

$$\sum_{a=1}^N \Delta\mathcal{F}_{AB}(a) = \Delta F_{AB} = \sum_{a=1}^N \frac{\left\langle \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} \left(e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right) \right\rangle}{\left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right\rangle} \Delta F_{AB}, \quad (76)$$

so dividing through by the free energy gives a set of weights

$$\sum_{a=1}^N \frac{\left\langle \frac{\Delta u_a}{\sum_{k=1}^N \Delta u_k} \left(e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right) \right\rangle}{\left\langle e^{-\beta \sum_{k=1}^N \Delta u_k} - 1 \right\rangle} = 1 \quad (77)$$

then the correct form of the weights are simply

$$w_a = \frac{\left\langle \frac{\Delta u_a}{\Delta U} \left(e^{-\beta \Delta U} - 1 \right) \right\rangle}{\left\langle e^{-\beta \Delta U} - 1 \right\rangle} \quad (78)$$

and the free energy change ΔF_{AB} can be calculated using standard techniques.

Molecular Dynamics Information

Caffeine Simulations

Lambda Schedule

Caffeine was run with the CHARMM36 forcefield in NAMD excluding scaled 1-4 interactions with scaling constant 1.0, PME electrostatics with a grid spacing of 1Å and a dielectric constant of 1.0. The TIP3P water model was used. Switching was used from 9Å till the cutoff of 11Å. The pair list distance was 12.5 Å and 20 steps per cycle were used.

Table S1: The 32-window λ schedule used for the caffeine FEP simulations.

λ -window	start	end	λ -window	start	end
1	0	0.01	17	0.19	0.21
2	0.01	0.02	18	0.21	0.22
3	0.02	0.03	19	0.22	0.23
4	0.03	0.04	20	0.23	0.24
5	0.04	0.05	21	0.24	0.26
6	0.05	0.06	22	0.26	0.28
7	0.06	0.07	23	0.28	0.3
8	0.07	0.08	24	0.3	0.32
9	0.08	0.09	25	0.32	0.38
10	0.09	0.1	26	0.38	0.48
11	0.1	0.115	27	0.48	0.6
12	0.115	0.13	28	0.6	0.8
13	0.13	0.145	29	0.8	0.89
14	0.145	0.16	30	0.89	0.95
15	0.16	0.175	31	0.95	0.975
16	0.175	0.19	32	0.975	1

The simulation box for the start of the NpT equilibration was a rhombic dodecahedron with cell vectors (31 0 0),(0 31 0),(15.5 15.5 21.920310216783) and origin (0,0,0).

A .dcd frame was saved every 2500 steps. Each timestep was 2fs, nonbonded terms were re-evaluated every step and electrostatic terms every 2 steps. All bonds were rigid. Langevin temperature control was used with a damping constant of 1 and a target temperature of 300 K.

For the NpT equilibration Langevin piston control was used with a target pressure of 1.01325 bar, and piston period decay and temperature of 100, 50 and 300 respectively.

The energy of the system for the BAR FEP calculation was output every 500 steps. An alchemical softening constant of 5 was used for the Lennard-Jones interactions. Each FEP window was run for 1000000 steps of which 100000 were equilibration steps. There were 32 λ -windows in the whole simulation shown in Table S1. To vary the electrostatics and LJ terms separately in one FEP simulation the NAMD controls alchElecLambdaStart=0.6 and alchVdwLambdaEnd=1.0 were set.

Simulations of HIV-1P

Table S2: Table of the 32 window λ schedule for the binding and unbinding of the Lennard-Jones (LJ) interactions. The 16 window λ schedule for the swapping of charges between the ligands.

λ -window LJ	start	end	λ -window LJ	start	end	λ -window Electrostatic	start	end
1	0.00	0.01	17	0.16	0.18	1	0.0000	0.0625
2	0.01	0.02	18	0.18	0.20	2	0.0625	0.1250
3	0.02	0.03	19	0.20	0.22	3	0.1250	0.1875
4	0.03	0.04	20	0.22	0.24	4	0.1875	0.2500
5	0.04	0.05	21	0.24	0.28	5	0.2500	0.3125
6	0.05	0.06	22	0.28	0.32	6	0.3125	0.3750
7	0.06	0.07	23	0.32	0.36	7	0.3750	0.4375
8	0.07	0.08	24	0.36	0.40	8	0.4375	0.5000
9	0.08	0.09	25	0.40	0.44	9	0.5000	0.5625
10	0.09	0.10	26	0.44	0.52	10	0.5625	0.6250
11	0.10	0.11	27	0.52	0.60	11	0.6250	0.6875
12	0.11	0.12	28	0.60	0.68	12	0.6875	0.7500
13	0.12	0.13	29	0.68	0.76	13	0.7500	0.8125
14	0.13	0.14	30	0.76	0.84	14	0.8125	0.8750
15	0.14	0.15	31	0.84	0.92	15	0.8750	0.9375
16	0.15	0.16	32	0.92	1.00	16	0.9375	1.0000

Binding and Unbinding the Lennard-Jones Terms

The binding and unbinding FEP simulations were run with the CHARMM36 forcefield in NAMD excluding scaled 1-4 interactions with scaling constant 1.0, PME electrostatics with a grid spacing of 1Å and a dielectric constant of 1.0. The TIP3P water model was used. Switching was used from 9Å till the cutoff of 11Å . The pair list distance was 12.5Å and 20 steps per cycle were used.

The simulation box for the start of the NpT equilibration was a rhombic dodecahedron with cell vectors (91,0,0),(0,91,0) and (45.5,45.5,64.3467170879758) and origin (0,0,0). For the NpT equilibration Langevin piston control was used with a target pressure of 1.01325 bar, and piston period decay and temperature of 100, 50 and 300 respectively.

A .dcd frame was saved every 50 steps. Each timestep was 2 fs, nonbonded terms were re-evaluated every step and electrostatic terms every 2 steps. All simulated bonds were rigid.

Langevin temperature control was used with a damping constant of 1 and a target temperature of 300 K. The energy of the system for the BAR FEP calculation was output every 250 steps. An alchemical softening constant of 5 was used for the Lennard-Jones interactions. Each FEP window was run for 50000 steps of which 10000 were equilibration steps. The lambda schedule had 32 windows as shown in Table S2

Swapping Charges

For the charge swapping stage the lambda schedule had 16 windows as shown in Table S2. All other parameters were the same as the turning on and off of the Lennard-Jones terms, as described above.

AFEP Calculations

All AFEP results for all systems were taken from 1800, 0.2 ps frames. The first 200 frames were treated as equilibration frames for each lambda window.